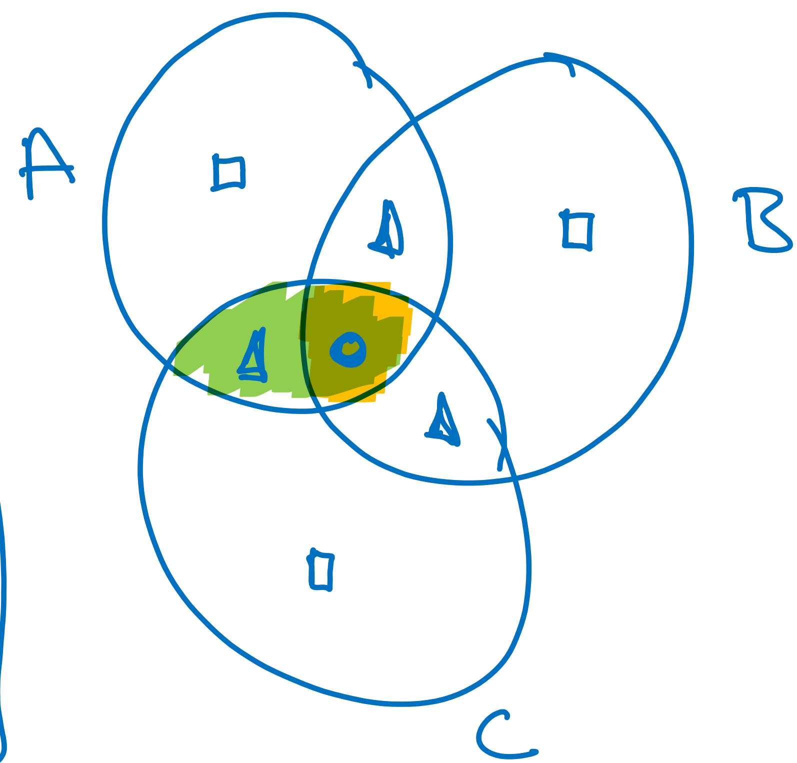
Lecture 21

I Inclusion-Exclusion

Recall: A, B finite, $|A \cup B| = |A| + |B| - |A \cap B|$.

3 sets: A, B, C.

$$\boxed{\begin{aligned} &|A \cup B \cup C| \\ &= |A| + |B| + |C| \\ &- |A \cap B| - |B \cap C| - |C \cap A| \\ &+ |A \cap B \cap C|. \end{aligned}}$$



Left hand Side

Right hand Side

II Review

□ : appears in 1 set $\qquad 1 \qquad 1$

△ : appear in 2 sets. $\qquad 1 \qquad 1+1-1=1$

○ : appears in $A\cap B\cap C$ $\qquad 1 \qquad 3-3+1=1$

Verifies ★.

Each term is counted the same number of times on both sides, So ★ holds.

(Q) Can you do n elts recursively?

Theorem: If $A_1,\dots,A_n$ are finite sets

$$\underbrace{|A_1\cup A_2\cup\dots\cup A_n|}_{\text{LHS}} = |A_1|+|A_2|+\dots+|A_n| \qquad \text{——— } n$$

$$-\,|A_1\cap A_2| - |A_1\cap A_3| - \dots \qquad \text{——— } \binom{n}{2}$$

$$+\,|A_1\cap A_2\cap A_3| + |A_1\cap A_2\cap A_4| + \dots \qquad \text{——— } \binom{n}{3}$$

$$\vdots \qquad\qquad \vdots$$

$$+\,(-1)^{n-1}\,|A_1\cap A_2\cap\dots\cap A_n|$$

$$= \sum_{k=1}^{n} (-1)^{k-1} \sum_{\substack{S \subseteq [n] \\ |S| = k}} \left| \bigcap_{i \in S} A_i \right|$$ RHS.

summation over S with these properties

where:

$$\bigcap_{i \in S} A_i = A_{i_1} \cap A_{i_2} \cap A_{i_3} \cdots \cap A_{i_k} \text{ where } S = \{i_1, \dots, i_k\}.$$

Proof: Let $A_1, \dots, A_n$ be arbitrary finite sets.

If $A_1 \cup \dots \cup A_n = \phi$ then $A_i = \phi$ for all i and the identity holds.

Let x be an arbitrary element of $A_1 \cup A_2 \cup \dots \cup A_n$.

x is counted once in the LHS.

Let $R = \{ i \in [n] : x \in A_i \}$. Let $t = |R|$.

x

A_1, A_3, A_n.

$R = \{1, 3\}$

The number of times x is counted on the RHS is:

$$\left(\#\text{ of } A_i \text{ containing } x\right) - \left(\#\text{ of pairs } A_i \cap A_j \ni x\right) + \left(\#\text{ of triples } A_i \cap A_j \cap A_k \ni x\right) \ldots$$

$$= |R| - \left(\#\text{ of 2 elt subsets of } R\right) + \left(\#\text{ of 3 element subsets of } R\right) \ldots\ldots$$

$$= t - \binom{t}{2} + \binom{t}{3} - \binom{t}{4} \ldots + (-1)^{t-1}\binom{t}{t}.$$

$= 1$ by the observation below.

Observe: $0 = (1+(-1))^t \underset{=}{\text{Binomial}} 1 - t + \binom{t}{2} - \binom{t}{3} \ldots + (-1)^t\binom{t}{t}$

Since every x is counted once on both sides, the identity holds. $\square$

ex: Suppose there are 180 ticked passengers with seat assignments on a flight. How many seating arrangements are there such that no passenger sits in their assigned seat.

$\boxed{\pi_i \neq i}$

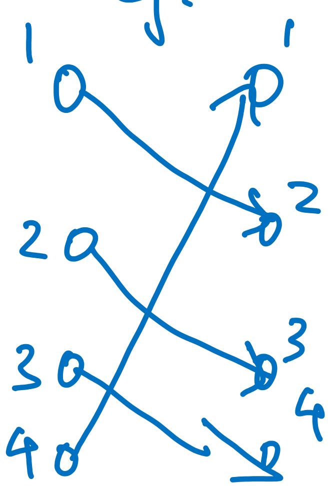
$n = 180$. π_i : seat for passenger i.

Defn: A permutation $(\pi_1, \_\_ \pi_n)$ of $\{1, \ldots, n\}$ is called a derangement if $\pi_i \neq i$ for all i.

An $i \in \{1, \ldots n\}$ is called a fixed point if $\pi_i = i$.

A derangement has no fixed points.

eg.



$$S_n = \{ (\pi_1, \_\_ \pi_n) : \text{permutation of } [n] \}.$$

$$D_n = \{ \pi \in S_n : \forall i \leq n \ \ \pi_i \neq i \}.$$

<u>Idea:</u> Let $A_1 = \{\pi \in S_n : \pi_1 = 1\} \subseteq S_n$

$A_2 = \{\pi \in S_n : \pi_2 = 2\}.$

$\vdots$

$A_n = \{\pi \in S_n : \pi_n = n\} \subseteq S_n$

Observe that $\boxed{D_n = S_n - A_1 \cup A_2 \cup A_3 \dots \cup A_n}$ complement.

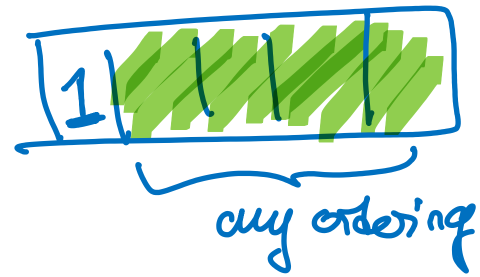
$|D_n| = \underbrace{|S_n|}_{n!} - \underbrace{|A_1 \cup \dots \cup A_n|}_{\text{can analyze using I.E.}}$ ← π with ≥ 1 fixed pt.

By Inclusion-Exclusion:

$$\underbrace{|A_1 \cup \dots \cup A_n|}_{\text{union}} = \sum_{k=1}^{n} (-1)^{k-1} \sum_{\substack{T \subseteq [n] \\ |T| = k}} \underbrace{\left| \bigcap_{i \in T} A_i \right|}_{\text{intersection}}.$$

Notice:

$A_1 = \{ \pi \in S_n : \pi_1 = 1 \}$.

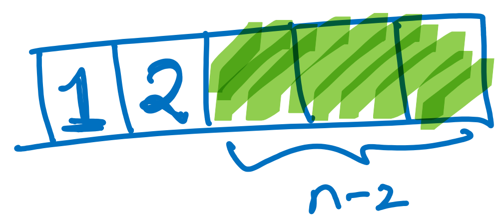


$|A_1| = (n-1)!$

In general,

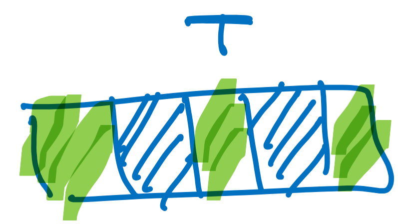
$|A_i| = (n-1)!$ for all i.

$|A_1 \cap A_2| = (n-2)!$



In general

$|A_i \cap A_j| = (n-2)!$ $\forall\, i \neq j$.



In general, if $|T| = k$, $\boxed{\left| \bigcap_{i \in S} A_i \right| = (n-k)!}$,

Since a $\pi \in \bigcap_{i \in S} A_i$ is uniquely specified by fixing $\pi_i = i$ for all $i \in T$ and choosing an arbitrary permutation of $[n]-T$.

Plugging this into I.E:

$$|A_1 \cup \dots \cup A_n| = \sum_{k=1}^{n} (-1)^{k-1} \sum_{\substack{T \subseteq [n] \\ |T|=k}} (n-k)!$$

only depends on $|T|$.

of $|T|=k$

$$= \sum_{k=1}^{n} (-1)^{k-1} \binom{n}{k} (n-k)!$$

$$= \sum_{k=1}^{n} (-1)^{k-1} \frac{n!}{\cancel{(n-k)!}\, k!} \cancel{(n-k)!}$$

$$\therefore \quad |D_n| = n! - n! \left(\frac{1}{1!} - \frac{1}{2!} + \frac{1}{3!} - \dots (-1)^{n-1} \frac{1}{n!} \right)$$

$$= n! \left(1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \dots (-1)^{n-1} \frac{1}{n!} \right)$$

22

$$e^{-1} = 1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} \cdots\cdots$$

$$\therefore \boxed{\lim_{n\to\infty} \frac{|D_n|}{n!} = \frac{1}{e}}$$

i.e. for large n, $\boxed{1/e \approx 37\%}$ fraction of π are derangements.

II Review

Induction

* What is n?
  What is $P(n)$?

* Base Case.

* Inductive Step
  Relate object of size $k+1$ to smaller

  k $\qquad$ $\leq k$ Strong Induction.

* Inductive loading: proving a stronger stmt can be easier.

Graph Theory

* Give everything a name
  - a cycle.
  - a cycle C.
  - a cycle $C = x_0, e_1, x_1, \ldots, x_k$.

* Use keywords (traverse, visit, concatenate..)

* Draw pictures, but be aware of generality.

* Delete edge/vertex/cycle to do induction.

* Pass to connected components.

* Big picture.

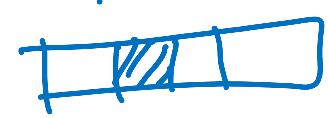
Counting

* Define a process for uniquely specifying objects.

* What is the bijection?

* Ask for justification of every $+, -, \times, \div$.

* Draw



(Q) Induction on vertices vs edges?

(A) Choose whichever yields a smaller graph which satisfies P(k), and allows you to get P(k+1) from P(k).

Given: All graph theory defns.

Can Cite: Any theorem/proposition proven in class.

→ State it, name.

(Not theorems on HW)

Difficulty ~ practice exam.

- Understand
- form a belief
- find strategy.
- try to prove.

Quiz this week: Ungraded, highly recommended.

Problem: Give a combinatorial proof that if $n \geq 1$, $k \leq n$

$$\sum_{k=0}^{n} k\binom{n}{k} = n \cdot 2^{n-1}.$$

product

sum

① Unpack defns.

$\binom{n}{k}$ ↳ # of k-element subsets of $[n]$.

LHS = RHS by counting the same thing in 2 ways.

what is that thing?

↳ reverse engineer from LHS/RHS.

LHS:

$$\boxed{\sum_{k=0}^{n} k\binom{n}{k}} \rightsquigarrow A_0 \cup A_1 \cup \dots \cup A_n$$

disjoint

s.t. $\boxed{|A_k| = k\binom{n}{k}.}$

If $k \geq 1$:

Consider the process

⓪ Choose k

$\binom{n}{k}$ ① Choose a k-subset $S \subseteq [n]$.

k ② Choose an element $x \in S$.

①,②: Thus uniquely specifies a k-subset S with a distinguished element $(S,x): x\in S$.

⓪,①,②: Uniquely specifies a nonempty subset $S \subseteq [n]$ and an element $x \in S$.

Other Process:

Choose $x \in [n]$ — n ways.

Choose $T \subseteq [n]-\{x\}$ — 2^{n-1}

Let $S = T \cup \{x\}$.

$$\therefore \sum_{k=0}^{n} k\binom{n}{k} = n\,2^{n-1}.$$